Omnibase Docs

Developers

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Current State of Project

asdfasdfasdfasdf

To do:

- Poner comentarios con qeu ID es cada motor FL, BL, FR, BR
- Implementar máquina de estados para que no se tenga que enviar y recibir absolutamente todos los valores (si no son muchos aun así jala bien pero pa q sea más mamon)??
- Hacer que BT_active sea 1 si está conectado y 2 si está activo para verlo en dashboard
- Hacer que la esp mande un heartbeat para corroborar que sigue conectado

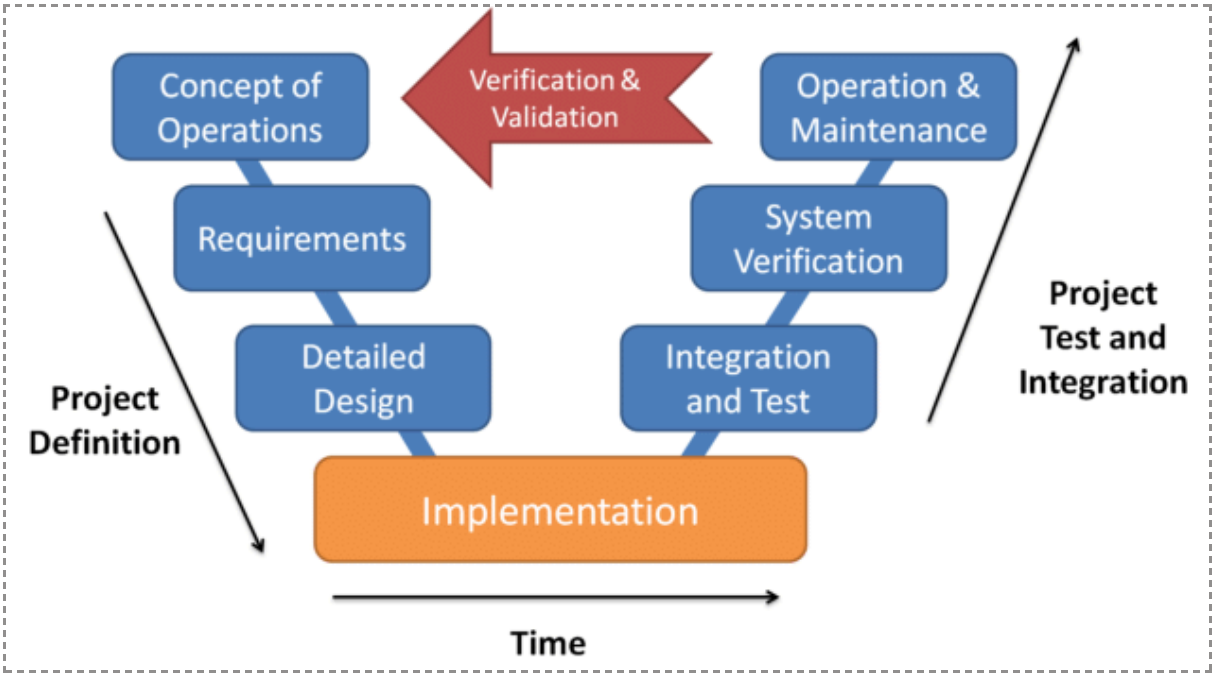
Index

PINOUT

PIN	Función
PB6	I2C_SCL (BNO055)
PB7	I2C_SDA (BNO055)
PD5	UART2_TX (ESP32)
PD6	UART2_RX (ESP32)
PD0	FDCAN1_RX (TJA1051)
PD1	FDCAN1_TX (TJA1051)
PD8 (USB port)	UART2_TX
PD9 (USB port)	UART2_RX

REQUERIMIENTOS DEL SISTEMA

Por Deivideich



Requerimientos del sistema

- Debe tener un MCU (STM32HX *o similar*)
- Debe tener un MCU auxiliar para comunicación inalámbrica (ej. ESP32)
    - Alternativamente un módulo de comunicación inalámbrica
    - Ej. Módulo RF24
- Todos los periféricos se conectan a esta placa. (XT60, salidas de drivers, entradas de alimentación, cables ethernet, comunicación)
- Botones de reset y otros básicos para debug
- Serial para comunicarse con la computadora principal, a decisión del diseño, UART, RJ45, etc.
- Sensores integrados
    - Ultrasónicos
    - Conexión a motores (drivers)
    - BNO/IMU
    - LIDAR (canal de coms)
    - Encoders
    - Mínimo dos bumpers
    - Al menos una entrada para botón (user button)
    - Mini fusibles para la STM
        - Revisar con la especificación de la protección integrada de overcurrent de STM
        - Polyfuse
- Salida de información por medio sonoro (buzzer)
- Monitoreo de otras placas:
    - Sensado de corriente global
    - Temperatura de componentes selectos

- Pines de comunicaciones
    - GPIO
    - Protocolos de comuniación
- STLInk
- ROHS

REQUERIMIENTOS DEL HARDWARE

Por David H
Requerimentos de hardware

- Uso de STM32H755ZIT6 para realizar control de los motores
- ESP32 para comunicación inalámbrica
- Mirco-usb para comunicación Compu – Placa
- IMU
- Probar interfaz ethernet
- Conectores JST o Molex-470531000
- Botones NA para user button y reset
- Con la compu pinricipal aún está pendiente comparar Micro-USB vs Ethernet
- Ultrasónicos
    - HC-04
- Conexiones a motores a través de interfaz CAN
- Drivers Odrive S1
    - Comunicación a través de FDCAN de stm32h7
    - Interfaz CAN MCP2551 (Como alternativa al FDCAN)

TUTORIAL

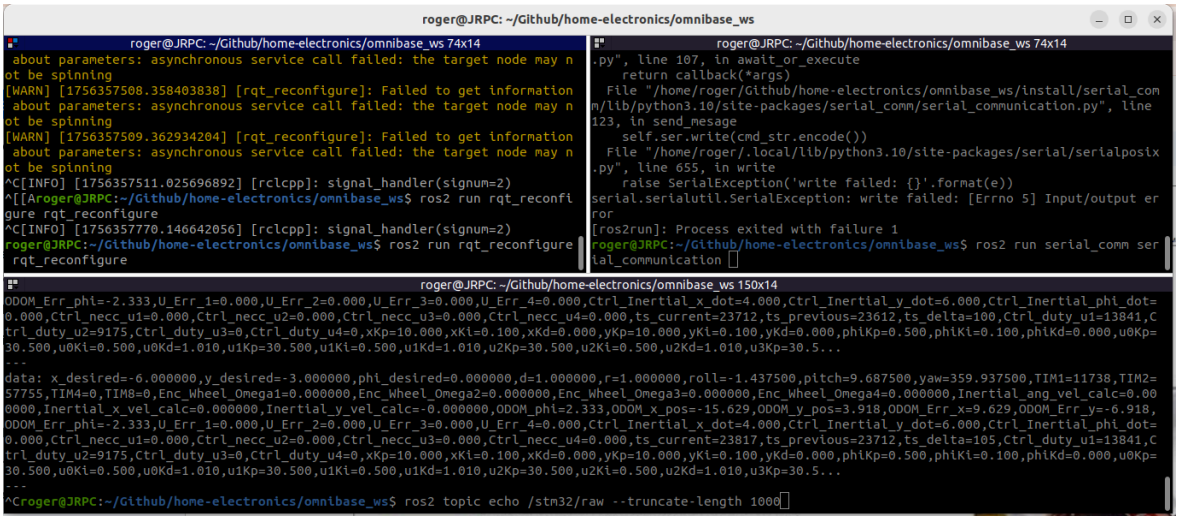
- Connect STM32H7 to Computer (laptop, rpi, jetson)
- If not already there download the github repo
  https://github.com/RoBorregos/home-custom-base/tree/firmware
- Within the repo go to the omnibase_ws folder, build it, and source it
- run the following command

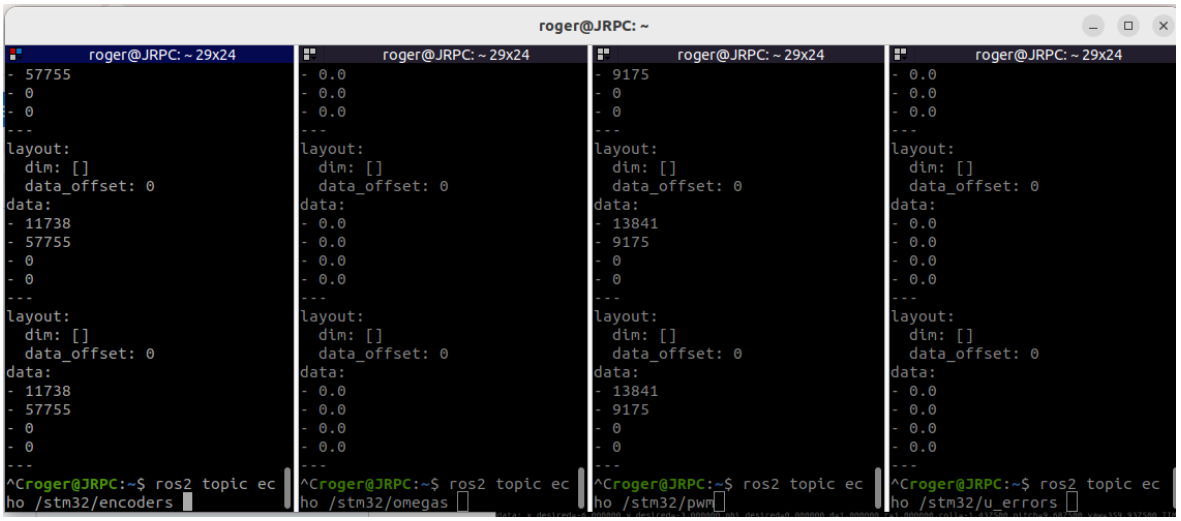
  ros2 run odrive_comm odrive_dashboard

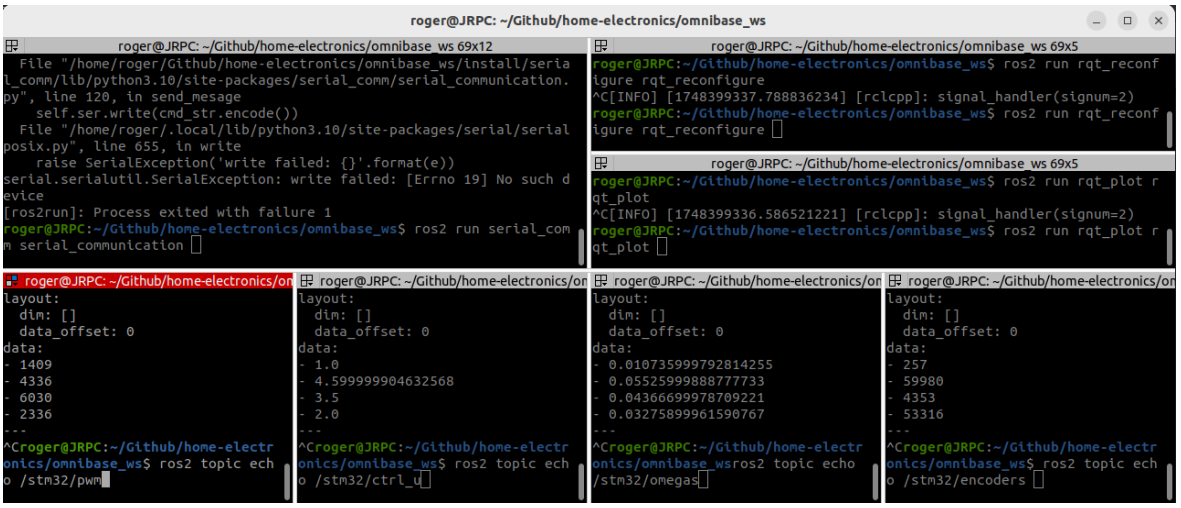
- The dashboard should be available on:

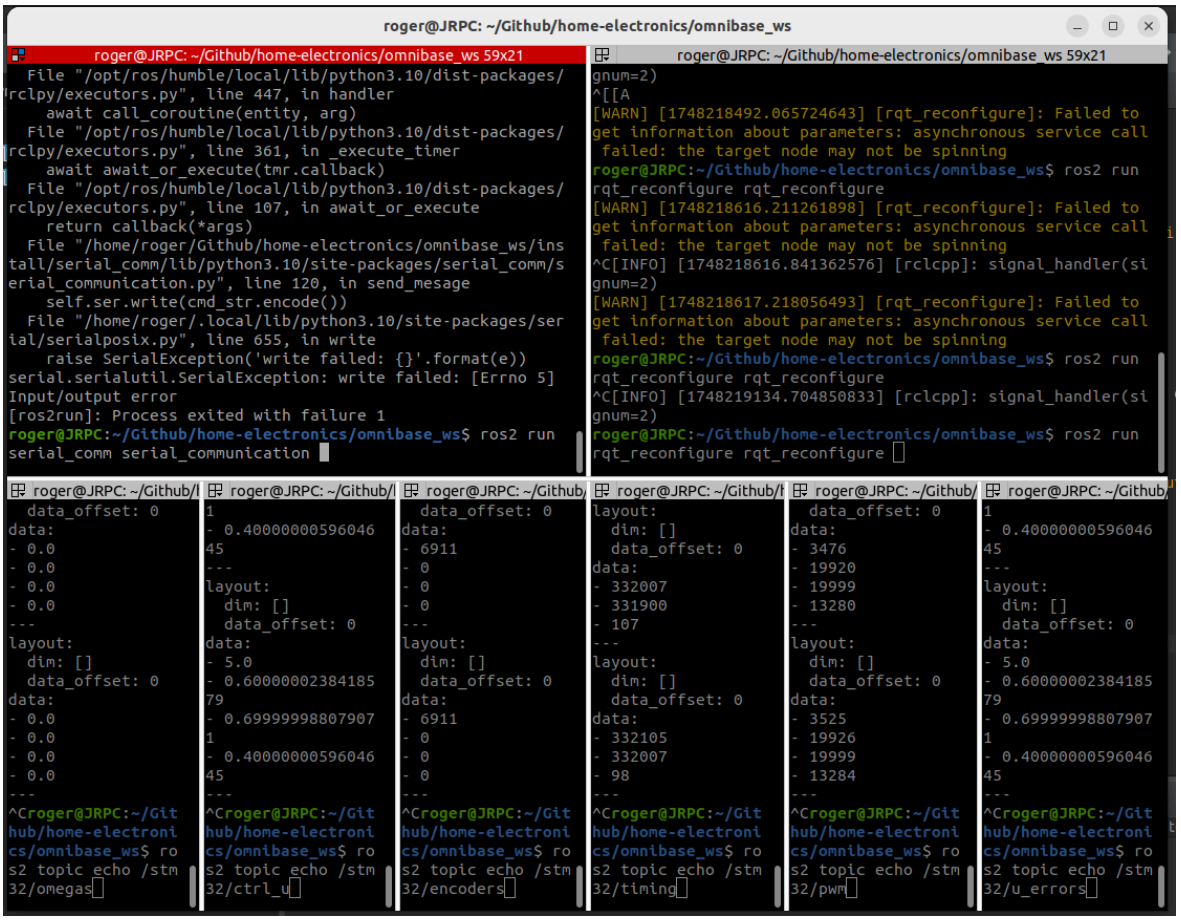
  http://localhost:5000/offline

ROS2 (PENDIENTE)






roger@JRPC: ~/Github/home-electronics/omnibase_ws
roger@JRPC: ~/Github/home-electronics/omnibase_ws 69x12
File "/home/roger/Github/home-electronics/omnibase_ws/install/serial_comm/lib/python3.10/site-packages/serial_comm/serial_communication.py", line 120, in send_mesage
self.ser.write(cmd_str.encode())
File "/home/roger/.local/lib/python3.10/site-packages/serial/serialposix.py", line 655, in write
raise SerialException('write failed: {}'.format(e))
serial.serialutil.SerialException: write failed: [Errno 19] No such device
[ros2run]: Process exited with failure 1
roger@JRPC:~/Github/home-electronics/omnibase_ws$ ros2 run serial_com m serial_communication
roger@JRPC: ~/Github/home-electronics/omnibase_ws 69x5
roger@JRPC:~/Github/home-electronics/omnibase_ws$ ros2 run rqt_reconf igure rqt_reconfigure
^C[INFO] [1748399337.788836234] [rclcpp]: signal_handler(signum=2)
roger@JRPC:~/Github/home-electronics/omnibase_ws$ ros2 run rqt_reconf igure rqt_reconfigure
roger@JRPC: ~/Github/home-electronics/omnibase_ws 69x5
roger@JRPC:~/Github/home-electronics/omnibase_ws$ ros2 run rqt_plot r qt_plot
^C[INFO] [1748399336.586521221] [rclcpp]: signal_handler(signum=2)
roger@JRPC:~/Github/home-electronics/omnibase_ws$ ros2 run rqt_plot r qt_plot
layout:
dim: []
data_offset: 0
data:
- 1409
- 4336
- 6030
- 2336
---
^Croger@JRPC:~/Github/home-electronics/omnibase_ws$ ros2 topic echo /stm32/pwm
layout:
dim: []
data_offset: 0
data:
- 1.0
- 4.599999904632568
- 3.5
- 2.0
---
^Croger@JRPC:~/Github/home-electronics/omnibase_ws$ ros2 topic echo /stm32/ctrl_u
layout:
dim: []
data_offset: 0
data:
- 0.010735999792814255
- 0.05525999888777733
- 0.04366699978709221
- 0.03275899961590767
---
^Croger@JRPC:~/Github/home-electronics/omnibase_wsros2 topic echo /stm32/omegas
layout:
dim: []
data_offset: 0
data:
- 257
- 59980
- 4353
- 53316
---
^Croger@JRPC:~/Github/home-electronics/omnibase_ws$ ros2 topic echo /stm32/encoders


roger@JRPC: ~/Github/home-electronics/omnibase_ws
roger@JRPC: ~/Github/home-electronics/omnibase_ws 59x21
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/executors.py", line 447, in handler
await call_coroutine(entity, arg)
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/executors.py", line 361, in _execute_timer
await await_or_execute(tmr.callback)
File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/executors.py", line 107, in await_or_execute
return callback(*args)
File "/home/roger/Github/home-electronics/omnibase_ws/install/serial_comm/lib/python3.10/site-packages/serial_comm/serial_communication.py", line 120, in send_mesage
self.ser.write(cmd_str.encode())
File "/home/roger/.local/lib/python3.10/site-packages/serial/serialposix.py", line 655, in write
raise SerialException('write failed: {}'.format(e))
serial.serialutil.SerialException: write failed: [Errno 5] Input/output error
[ros2run]: Process exited with failure 1
roger@JRPC:~/Github/home-electronics/omnibase_ws$ ros2 run serial_comm serial_communication
roger@JRPC: ~/Github/home-electronics/omnibase_ws 59x21
gnum=2)
^[[A
[WARN] [1748218492.065724643] [rqt_reconfigure]: Failed to get information about parameters: asynchronous service call failed: the target node may not be spinning
roger@JRPC:~/Github/home-electronics/omnibase_ws$ ros2 run rqt_reconfigure rqt_reconfigure
[WARN] [1748218616.211261898] [rqt_reconfigure]: Failed to get information about parameters: asynchronous service call failed: the target node may not be spinning
^C[INFO] [1748218616.841362576] [rclcpp]: signal_handler(signum=2)
[WARN] [1748218617.218056493] [rqt_reconfigure]: Failed to get information about parameters: asynchronous service call failed: the target node may not be spinning
roger@JRPC:~/Github/home-electronics/omnibase_ws$ ros2 run rqt_reconfigure rqt_reconfigure
^C[INFO] [1748219134.704850833] [rclcpp]: signal_handler(signum=2)
roger@JRPC:~/Github/home-electronics/omnibase_ws$ ros2 run rqt_reconfigure rqt_reconfigure
data_offset: 0
data:
- 0.0
- 0.0
- 0.0
- 0.0
---
layout:
dim: []
data_offset: 0
data:
- 0.0
- 0.0
- 0.0
- 0.0
---
^Croger@JRPC:~/Github/home-electronics/omnibase_ws$ ros2 topic echo /stm32/omegas
1
- 0.40000000596046
45
---
layout:
dim: []
data_offset: 0
data:
- 5.0
- 0.60000002384185
79
- 0.69999998807907
1
- 0.40000000596046
45
---
^Croger@JRPC:~/Github/home-electronics/omnibase_ws$ ros2 topic echo /stm32/ctrl_u
data_offset: 0
data:
- 6911
- 0
- 0
- 0
---
layout:
dim: []
data_offset: 0
data:
- 6911
- 0
- 0
- 0
---
^Croger@JRPC:~/Github/home-electronics/omnibase_ws$ ros2 topic echo /stm32/encoders
layout:
dim: []
data_offset: 0
data:
- 332007
- 331900
- 107
---
layout:
dim: []
data_offset: 0
data:
- 332105
- 332007
- 98
---
^Croger@JRPC:~/Github/home-electronics/omnibase_ws$ ros2 topic echo /stm32/timing
data_offset: 0
data:
- 3476
- 19920
- 19999
- 13280
---
layout:
dim: []
data_offset: 0
data:
- 3525
- 19926
- 19999
- 13284
---
^Croger@JRPC:~/Github/home-electronics/omnibase_ws$ ros2 topic echo /stm32/pwm
1
- 0.40000000596046
45
---
layout:
dim: []
data_offset: 0
data:
- 5.0
- 0.60000002384185
79
- 0.69999998807907
1
- 0.40000000596046
45
---
^Croger@JRPC:~/Github/home-electronics/omnibase_ws$ ros2 topic echo /stm32/u_errors

Odrive

Descripción

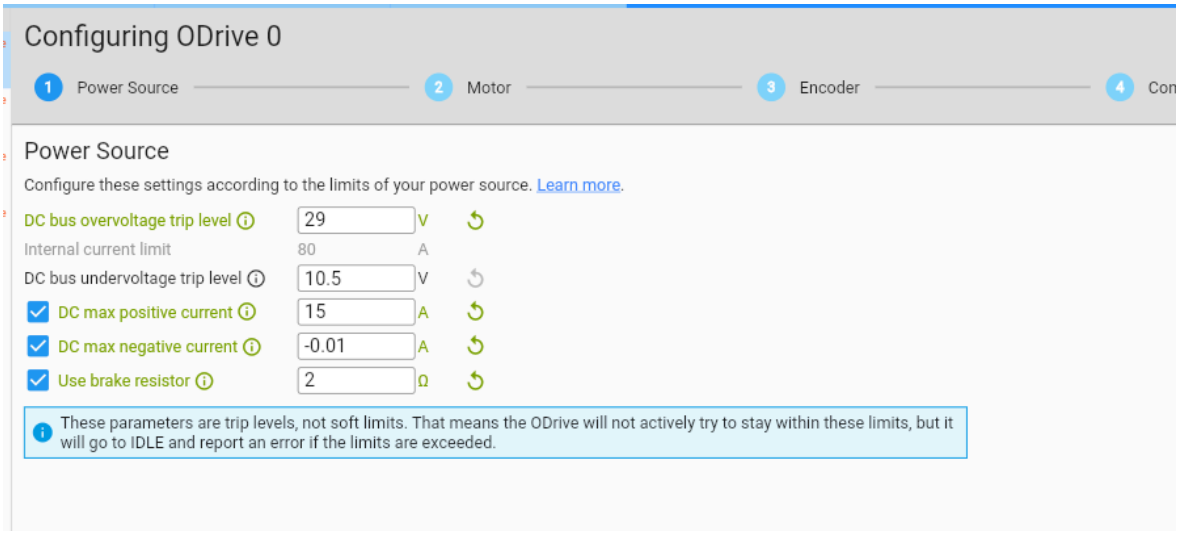
- Pay to win de control de brushless motor
- Se usa el modelo ODrive S1 (datasheet)
- Encoder incluido
- 12 V DC **IMPORTANTE: POLARIDAD NO INTERCAMBIABLE**

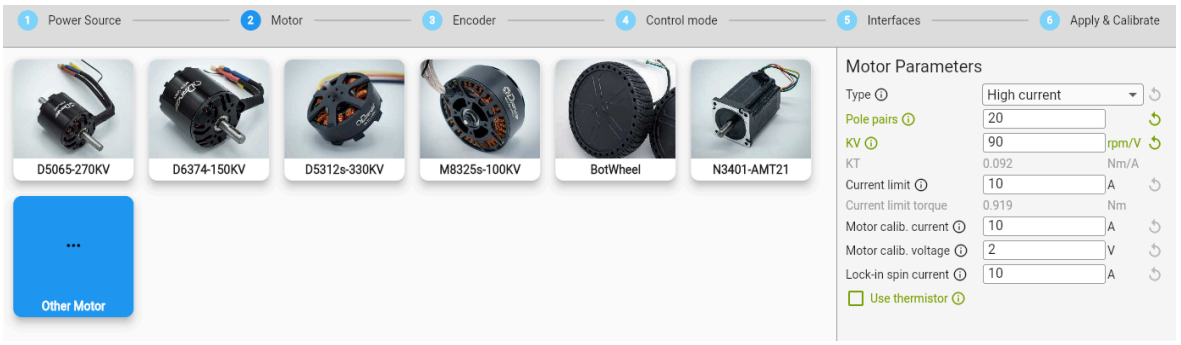
Configuración

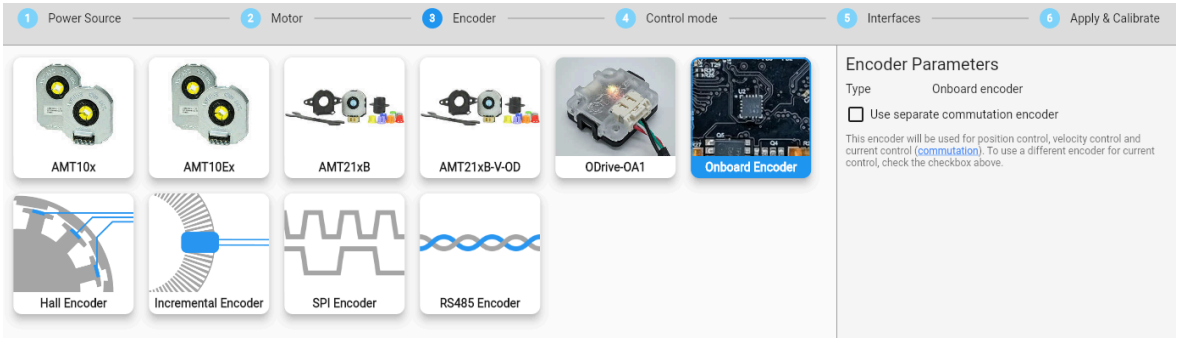
Para configurar el Odrive conectarlo a una computadora y configurarlo desde la GUI https://gui.odriverobotics.com/configuration:
(Nota: en teoría también se puede a través de CAN)

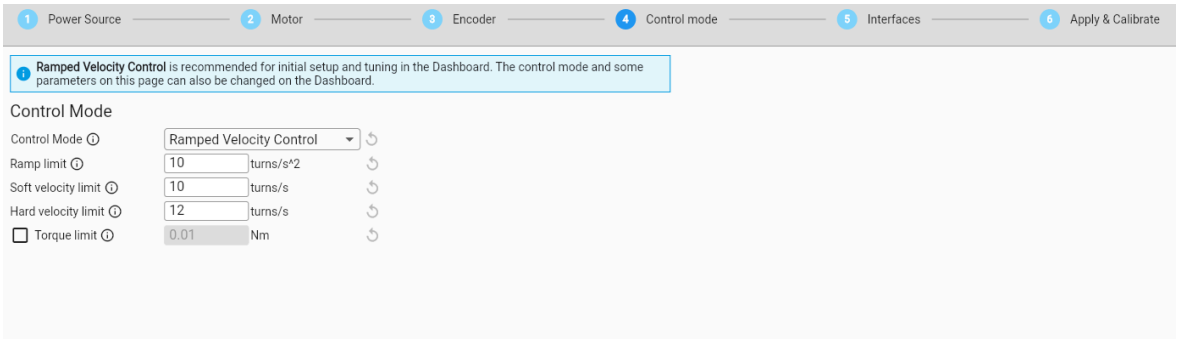
Troubleshooting: https://docs.odriverobotics.com/v/latest/interfaces/gui.html

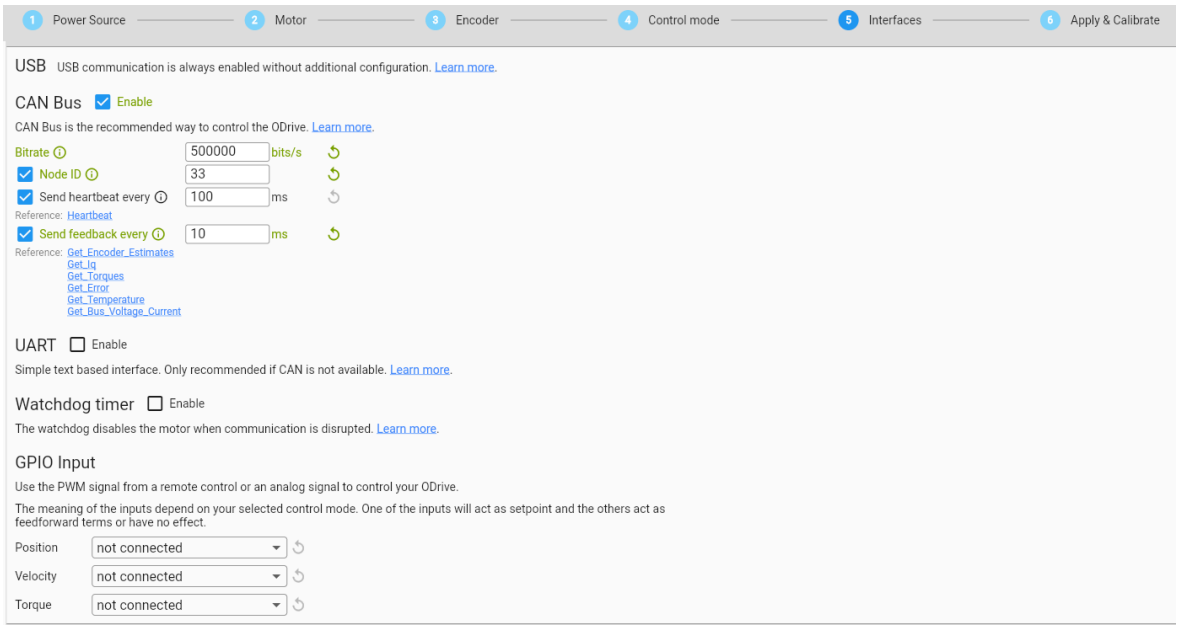
Configuración actual

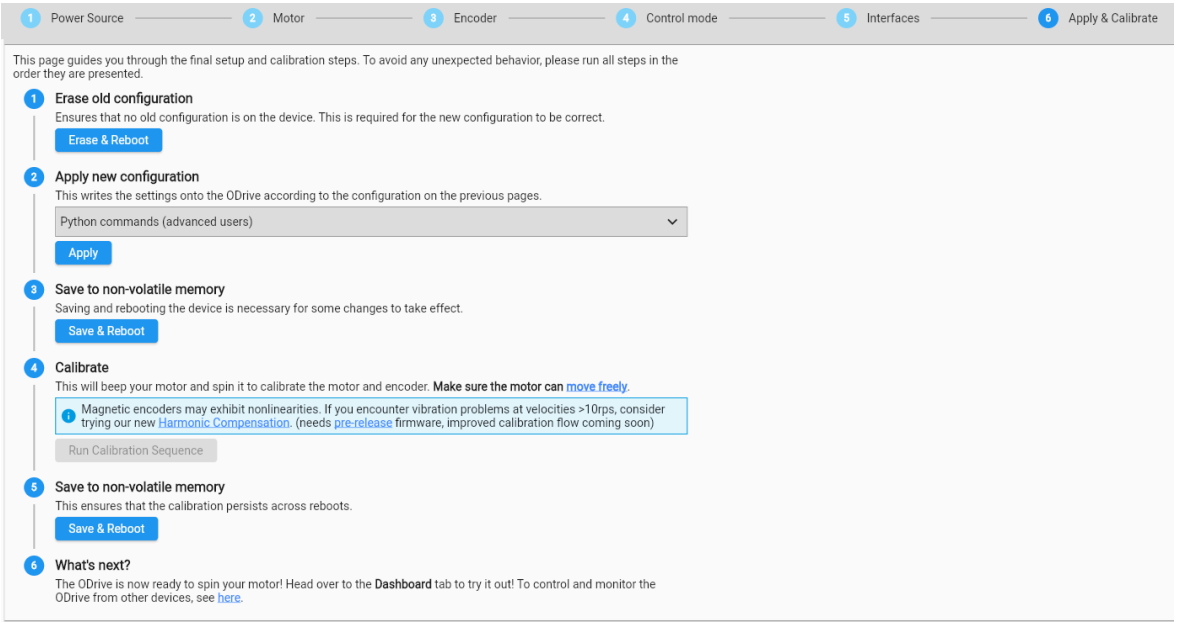





1 Power Source
2 Motor
3 Encoder
4 Control mode
5 Interfaces
6 Apply & Calibrate
AMT10x
AMT10Ex
AMT21xB
AMT21xB-V-OD
ODrive-OA1
Onboard Encoder
Hall Encoder
Incremental Encoder
SPI Encoder
RS485 Encoder
Encoder Parameters
Type Onboard encoder
Use separate commutation encoder
This encoder will be used for position control, velocity control and current control (commutation). To use a different encoder for current control, check the checkbox above.


1 Power Source
2 Motor
3 Encoder
4 Control mode
5 Interfaces
6 Apply & Calibrate
Ramped Velocity Control is recommended for initial setup and tuning in the Dashboard. The control mode and some parameters on this page can also be changed on the Dashboard.
Control Mode
Control Mode Ramped Velocity Control
Ramp limit 10 turns/s^2
Soft velocity limit 10 turns/s
Hard velocity limit 12 turns/s
Torque limit 0.01 Nm

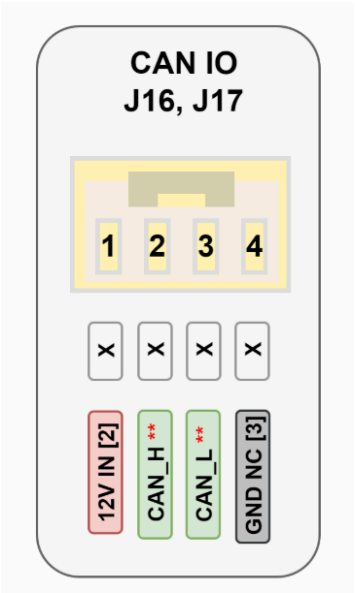

1 Power Source
2 Motor
3 Encoder
4 Control mode
5 Interfaces
6 Apply & Calibrate
USB USB communication is always enabled without additional configuration. Learn more.
CAN Bus Enable
CAN Bus is the recommended way to control the ODrive. Learn more.
Bitrate 500000 bits/s
Node ID 33
Send heartbeat every 100 ms
Reference: Heartbeat
Send feedback every 10 ms
Reference: Get_Encoder_Estimates
Get_Iq
Get_Torques
Get_Error
Get_Temperature
Get_Bus_Voltage_Current
UART Enable
Simple text based interface. Only recommended if CAN is not available. Learn more.
Watchdog timer Enable
The watchdog disables the motor when communication is disrupted. Learn more.
GPIO Input
Use the PWM signal from a remote control or an analog signal to control your ODrive.
The meaning of the inputs depend on your selected control mode. One of the inputs will act as setpoint and the others act as feedforward terms or have no effect.
Position not connected
Velocity not connected
Torque not connected

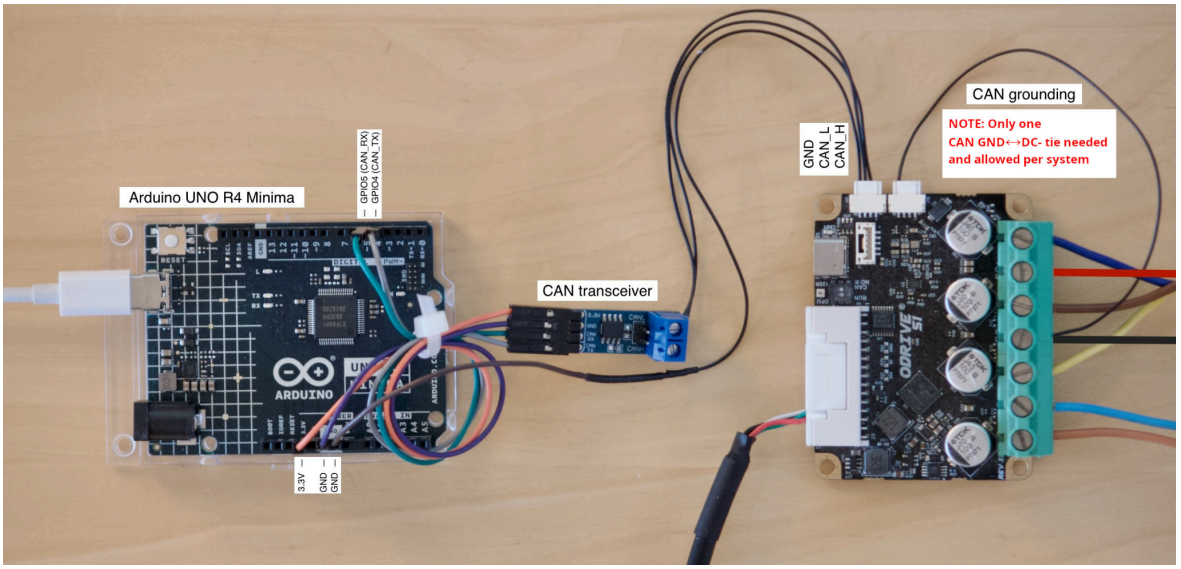


Si se cambia la configuración debes elección cada uno de los botones de esta sección en orden. La opción “Run Calibration Sequence” solo es posible al estar conectado a los 12V DC necesarios, consume ~1.3A al calibrar.

Odrive CAN

BITRATE En configuración actual 500kBs





Para conectar el odrive a CAN es necesario conectar los pines CAN_H y CAN_L a los correspondientes en el bus CAN y GND al GND de los microcontroladores.

Es necesario cambiar el Axis_State según el modo de control (checar documentación, no entendí bien todavia). Esto se puede hacer desde el gui en línea picandole al botón de power.

Base para implementacion de funciones de control/configuración del odrive por CAN:
https://github.com/siddarthiyer/ODrive-STM32-CAN-Driver/tree/main

https://docs.odriverobotics.com/v/latest/manual/can-protocol.html
https://docs.odriverobotics.com/v/latest/guides/can-guide.html
https://gui.odriverobotics.com/dashboard
https://docs.odriverobotics.com/v/latest/guides/odrivetool-setup.html
https://docs.odriverobotics.com/v/latest/guides/getting-started.html
https://docs.odriverobotics.com/v/latest/interfaces/gui.html

CANABLE

On linux

Install can-utils package:
sudo apt update
sudo apt install can-utils

Check that slcand is installed:
dpkg -L can-utils | grep slcand

Check what port the canable was connected to:

ls /dev/ttyACM /dev/ttyUSB* 2>/dev/null*

Enable that port as a can# intergace (-s6 significa 500kbit):

sudo slcand -o -c -s6 /dev/ttyACM1 can0
sudo ip link set up can0 txqueuelen 1000

More documentation: https://canable.io/getting-started.html#cangaroo

Example Commands:

- Create CAN interface from CANable (serial to SocketCAN):
  sudo slcand -o -c -s6 /dev/ttyACM0 can0
- Bring the CAN interface up (activate it):
  sudo ifconfig can0 up
- Increase transmit queue length to avoid dropped frames:
  sudo ifconfig can0 txqueuelen 1000
- Send a CAN frame with ID 0x999 and payload 0xDEADBEEF:
  cansend can0 999#DEADBEEF
- Dump all CAN traffic from can0 in real time:
  candump can0
- Show CAN bus load percentage at 500 kbit/s:
  canbusload can0@500000
- Sniff CAN traffic in top-style view (grouped by ID, highlights changes):
  cansniffer can0
- Generate fixed CAN messages with 8-byte payload 11 22 33 44 DE AD BE EF:
  cangen can0 -D 11223344DEADBEEF -L 8

MCP2515

Driver q falta checar:
https://github.com/siddarthiyer/ODrive-STM32-CAN-Driver/tree/main

Librería del MCP2515 usada:
https://github.com/thumptech/STM32-MCP2515?tab=readme-ov-file

- Se usaron los archivos mcp2515.c, mcp2515.h y mcp2515_consts.h además de incluir el can.h de la librería de arduino.
  https://github.com/autowp/arduino-mcp2515?tab=readme-ov-file

FD CAN (with TJA1051 PENDIENTE PROBAR)

Basic Parameters	
Frame Format	Classic mode
Mode	Normal mode
Auto Retransmission	Disable
Transmit Pause	Disable
Protocol Exception	Enable
Nominal Sync Jump Width	8
Data Prescaler	1
Data Sync Jump Width	1
Data Time Seg1	1
Data Time Seg2	1
Message Ram Offset	0
Std Filters Nbr	1
Ext Filters Nbr	0
Rx Fifo0 Elmts Nbr	1
Rx Fifo0 Elmt Size	8 bytes data field
Rx Fifo1 Elmts Nbr	0
Rx Fifo1 Elmt Size	8 bytes data field
Rx Buffers Nbr	0
Rx Buffer Size	8 bytes data field
Tx Events Nbr	0
Tx Buffers Nbr	0
Tx Fifo Queue Elmts Nbr	1
Tx Fifo Queue Mode	FIFO mode
Tx Elmt Size	8 bytes data field
Clock Calibration Unit	
Clock Calibration	Disable
Bit Timings Parameters	
Nominal Prescaler	2
Nominal Time Quantum	50.0 ns
Nominal Time Seg1	31
Nominal Time Seg2	8
Nominal Time for one Bit	2000 ns
Nominal Baud Rate	500000 bit/s

FDCAN in STM32 || LoopBack Mode

```
if (HAL_FDCAN_GetRxMessage(&hfdcan1, FDCAN_RX_FIFO0, &RxHeader, RxData) == HAL_OK) {
            if (RxHeader.IdType == FDCAN_EXTENDED_ID) {
                printf("Extended ID: 0x%lx\n\r", RxHeader.Identifier);
            } else {
                printf("Standard ID: 0x%lx\n\r", RxHeader.Identifier);
            }
```

```
            dataLength = (RxHeader.DataLength >> 16) & 0x0F; // Extract
DLC (0-8 bytes)
            for (int i = 0; i < (dataLength); i++) {
                printf("0x%02X ", RxData[i]);
            }
            printf("Message received successfully\n\r");
        } else {
            printf("No message received\n\r");
        }
            HAL_Delay(100); // Avoid tight loop
    if (HAL_FDCAN_AddMessageToTxFifoQ(&hfdcan1, &TxHeader, TxData) ==
HAL_OK) {
         printf("Message sent successfully\n\r");
    } else {
        printf("Failed to send message\n\r");
    }
       if (HAL_FDCAN_GetRxMessage(&hfdcan1, FDCAN_RX_FIFO0, &RxHeader,
RxData) == HAL_OK) {
            uint8_t dataLength = (RxHeader.DataLength & 0x0F); // Extract
DLC
            if (dataLength > 0) {
                printf("Received ID: 0x%lx\n\r", RxHeader.Identifier);
                printf("Data: ");
                for (int i = 0; i < dataLength; i++) {
                    printf("0x%02X ", RxData[i]);
                }
                printf("\n\r");
            } else {
                printf("Received message with no data.\n\r");
            }
        } else {
            printf("No message received.\n\r");
        }
```

```
static void MX_FDCAN1_Init(void)
{
 /* USER CODE BEGIN FDCAN1_Init 0 */
 /* USER CODE END FDCAN1_Init 0 */
 /* USER CODE BEGIN FDCAN1_Init 1 */
 /* USER CODE END FDCAN1_Init 1 */
 hfdcan1.Instance = FDCAN1;
 hfdcan1.Init.FrameFormat = FDCAN_FRAME_CLASSIC;
 hfdcan1.Init.Mode = FDCAN_MODE_NORMAL;
 hfdcan1.Init.AutoRetransmission = DISABLE;
 hfdcan1.Init.TransmitPause = DISABLE;
 hfdcan1.Init.ProtocolException = ENABLE;
 hfdcan1.Init.NominalPrescaler = 2;
 hfdcan1.Init.NominalSyncJumpWidth = 8;
 hfdcan1.Init.NominalTimeSeg1 = 31 ;
 hfdcan1.Init.NominalTimeSeg2 = 8;
 hfdcan1.Init.DataPrescaler = 1;
 hfdcan1.Init.DataSyncJumpWidth = 1;
```

```
 hfdcan1.Init.DataTimeSeg1 = 1;
 hfdcan1.Init.DataTimeSeg2 = 1;
 hfdcan1.Init.MessageRAMOffset = 0;
 hfdcan1.Init.StdFiltersNbr = 1;
 hfdcan1.Init.ExtFiltersNbr = 0;
 hfdcan1.Init.RxFifo0ElmtsNbr = 1;
 hfdcan1.Init.RxFifo0ElmtSize = FDCAN_DATA_BYTES_8;
 hfdcan1.Init.RxFifo1ElmtsNbr = 0;
 hfdcan1.Init.RxFifo1ElmtSize = FDCAN_DATA_BYTES_8;
 hfdcan1.Init.RxBuffersNbr = 0;
 hfdcan1.Init.RxBufferSize = FDCAN_DATA_BYTES_8;
 hfdcan1.Init.TxEventsNbr = 0;
 hfdcan1.Init.TxBuffersNbr = 0;
 hfdcan1.Init.TxFifoQueueElmtsNbr = 1;
 hfdcan1.Init.TxFifoQueueMode = FDCAN_TX_FIFO_OPERATION;
 hfdcan1.Init.TxElmtSize = FDCAN_DATA_BYTES_8;
 if (HAL_FDCAN_Init(&hfdcan1) != HAL_OK)
 {
   Error_Handler();
 }
 /* USER CODE BEGIN FDCAN1_Init 2 */
 /*AAO+*/
  /* Configure Rx filter */
 sFilterConfig.IdType = FDCAN_EXTENDED_ID;             // Set to Extended ID
 sFilterConfig.FilterIndex = 0;                        // Filter index
 sFilterConfig.FilterType = FDCAN_FILTER_MASK;         // Filter type (Mask
mode)
 sFilterConfig.FilterConfig = FDCAN_FILTER_TO_RXFIFO0;// Direct matching
frames to FIFO 0
 sFilterConfig.FilterID1 = 0x18FF1220; // Base ID
 sFilterConfig.FilterID2 = 0xFFFFFFF0; // Mask to allow only last 2 bits to
vary
   /* Configure global filter to reject all non-matching frames */
 HAL_FDCAN_ConfigGlobalFilter(&hfdcan1,
        FDCAN_ACCEPT_IN_RX_FIFO0,  // Accept standard messages in FIFO0
(if needed)
        FDCAN_ACCEPT_IN_RX_FIFO0,  // Accept extended messages in FIFO0
     FDCAN_REJECT_REMOTE,         // Reject remote frames
     FDCAN_REJECT_REMOTE);        // Reject remote frames
   if (HAL_FDCAN_ConfigFilter(&hfdcan1, &sFilterConfig) != HAL_OK)
     {
        /* Filter configuration Error */
        Error_Handler();
     }
    /* Start the FDCAN module */
   if (HAL_FDCAN_Start(&hfdcan1) != HAL_OK) {
     }
        /* Start Error */
   if (HAL_FDCAN_ActivateNotification(&hfdcan1,
FDCAN_IT_RX_FIFO0_NEW_MESSAGE, 0) != HAL_OK) {
     }
        /* Notification Error */
    /* Configure Tx buffer message */
```

```
  TxHeader.Identifier = 0x18FF1235;           // Example Extended ID
  TxHeader.IdType = FDCAN_EXTENDED_ID;      // Use Extended ID
  TxHeader.TxFrameType = FDCAN_DATA_FRAME; // Set as a data frame
  TxHeader.DataLength = FDCAN_DLC_BYTES_8; // 8-byte data length
  TxHeader.ErrorStateIndicator = FDCAN_ESI_ACTIVE;
  TxHeader.BitRateSwitch = FDCAN_BRS_OFF;
  TxHeader.FDFormat = FDCAN_CLASSIC_CAN;    // Use classic CAN
  TxHeader.TxEventFifoControl = FDCAN_NO_TX_EVENTS;
  TxHeader.MessageMarker = 0x00;
 /*AAO-*/
   /* USER CODE END WHILE */
/* USER CODE END FDCAN1_Init 2 */
}
```

RTOS

ODriveCanTask

- Read & send Odrives status odometry and status
    - Odometry
        - Twist with FK function
        - Position: integrating twist
    - Read periodic ODrive CAN status msgs
        - Send with Queue: CAN_2_UTX
- Process command from UartRX task
    - Receiving queue: URX_2_CAN
    - Using odrive.c functions:
        - Velocity CMD: Calculate required vel per motor
            - IK function
        - Config CMD: Send Odrive CAN Ctrl Msg

UartRxTask

- Rx data
- Parse
  ↳ URX → CTRL
  ↳ URX → ODrives
  ↳ URX → UTX

UartTxTask

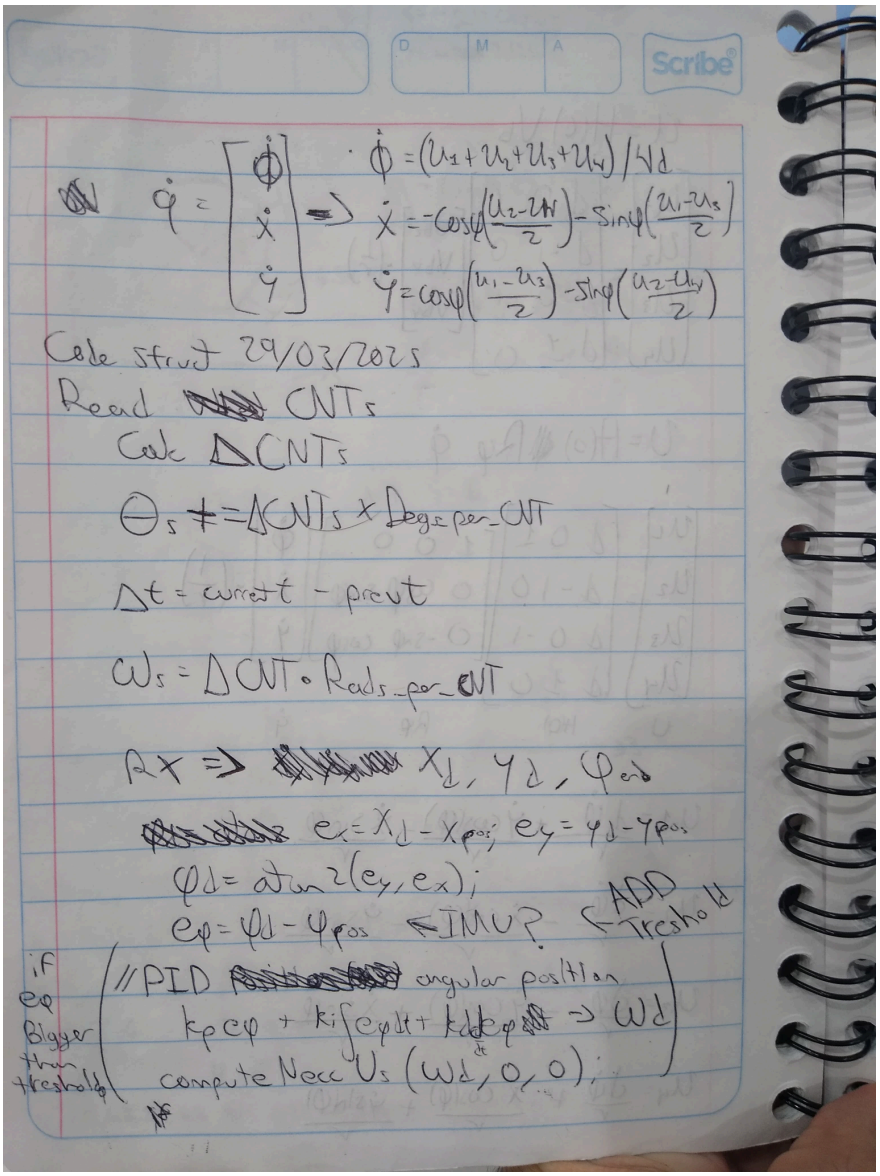
- Build msg
  ↳ UEX → UTX
  ↳ CTRL → UTX
  ↳ ODrive → UTX
- Tx data

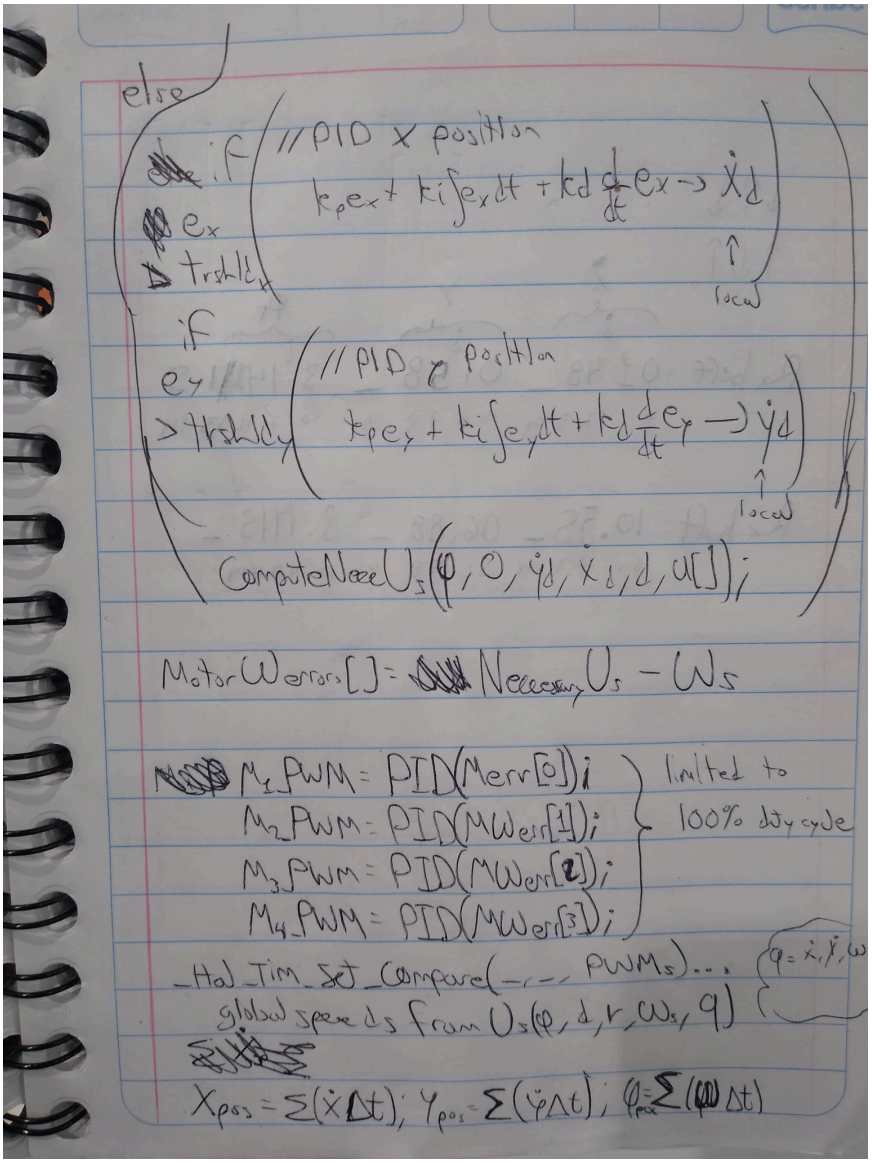
ROS2 SerialComm Node

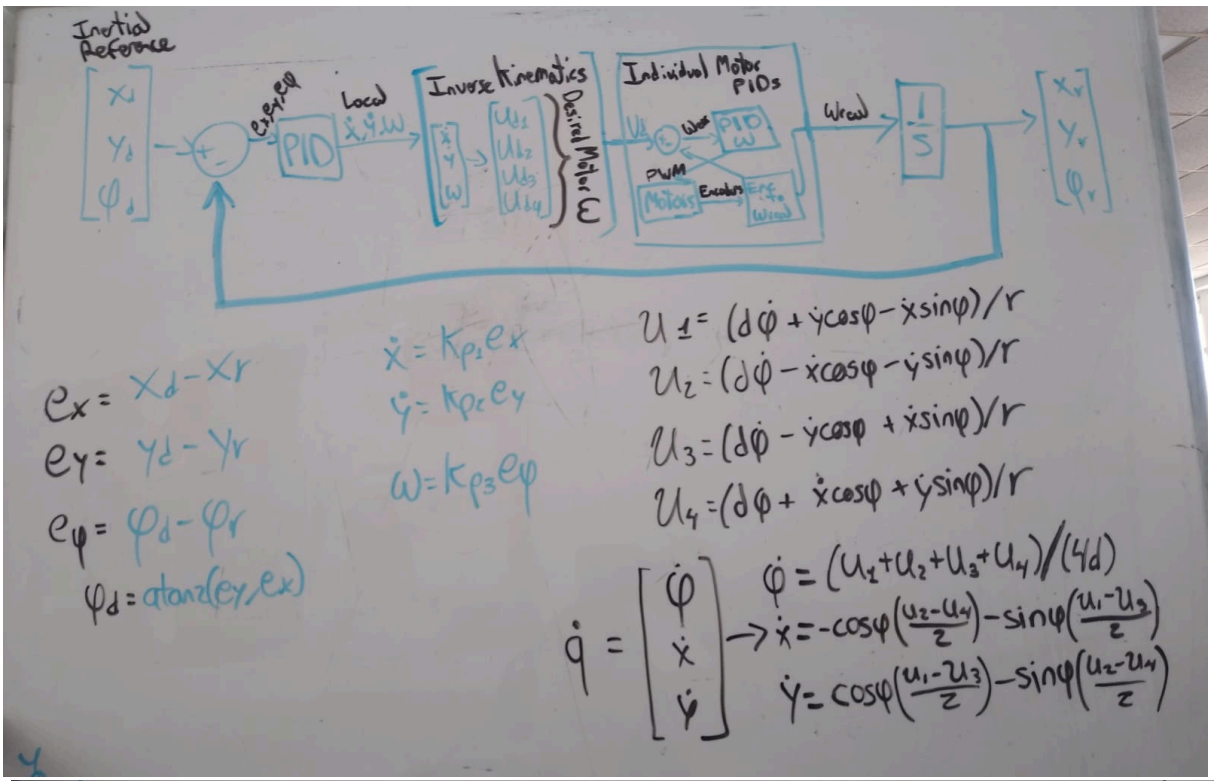
- **UartRX:** Status / Odom / Sensor Data
  ↳ Pub topics
- **UartTX:** Twist msg / Config
  ↳ Sub to Twist topic
  ↳ Sub to Config topic

Code Block Diagram (PENDIENTE)

OmniBasePositionCTRL_Attempt1 ← NOT YET TESTED




else
if
e_x
> threshold
// PID x position
$k_p e_x + k_i \int e_x dt + k_d \frac{d}{dt} e_x \rightarrow \dot{x}_d$
local
if
e_y
> threshold
// PID y position
$k_p e_y + k_i \int e_y dt + k_d \frac{d}{dt} e_y \rightarrow \dot{y}_d$
local
ComputeNeededU_s(φ, 0, ẏ_d, ẋ_d, d, u[]);
MotorWerrors[] = NeededU_s − W_s
M1_PWM = PID(Merr[0]);
M2_PWM = PID(MWerr[1]);
M3_PWM = PID(MWerr[2]);
M4_PWM = PID(MWerr[3]);
limited to 100% duty cycle
_Hal_Tim_Set_Compare(..., PWM_s)...
global speeds from U_s(φ, d, r, W_s, q)
q = ẋ, ẏ, ω
$X_{pos} = \Sigma(\dot{x}\Delta t)$; $Y_{pos} = \Sigma(\dot{y}\Delta t)$; $\varphi = \Sigma(\omega \Delta t)$



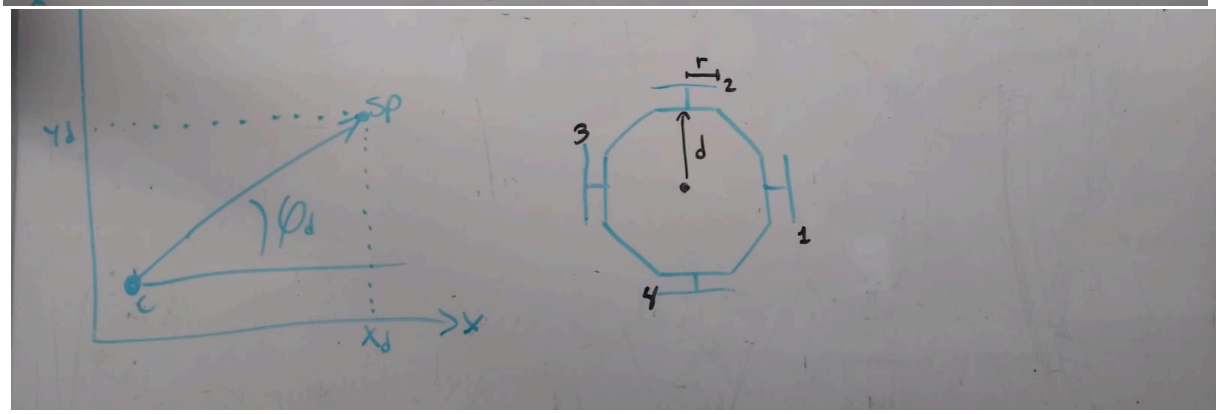
$$e_x = x_d - x_r$$
$$e_y = y_d - y_r$$
$$e_\varphi = \varphi_d - \varphi_r$$
$$\varphi_d = \text{atan2}(e_y, e_x)$$

$$\dot{x} = k_{p_1} e_x$$
$$\dot{y} = k_{p_2} e_y$$
$$\omega = k_{p_3} e_\varphi$$

$$u_1 = (d\dot{\varphi} + \dot{y}\cos\varphi - \dot{x}\sin\varphi)/r$$
$$u_2 = (d\dot{\varphi} - \dot{x}\cos\varphi - \dot{y}\sin\varphi)/r$$
$$u_3 = (d\dot{\varphi} - \dot{y}\cos\varphi + \dot{x}\sin\varphi)/r$$
$$u_4 = (d\dot{\varphi} + \dot{x}\cos\varphi + \dot{y}\sin\varphi)/r$$

$$\dot{q} = \begin{bmatrix} \dot{\varphi} \\ \dot{x} \\ \dot{y} \end{bmatrix}$$

$$\dot{\varphi} = (u_1 + u_2 + u_3 + u_4)/(4d)$$
$$\dot{x} = -\cos\varphi\left(\frac{u_2 - u_4}{2}\right) - \sin\varphi\left(\frac{u_1 - u_3}{2}\right)$$
$$\dot{y} = \cos\varphi\left(\frac{u_1 - u_3}{2}\right) - \sin\varphi\left(\frac{u_2 - u_4}{2}\right)$$



BNO055

Funcionó la BNO055 con la STM32H755 utilizando los archivos bno055.c, bno055.h y bno055_stm32.h de la siguiente librería:
https://github.com/ivyknob/bno055_stm32/tree/master

Fue necesario cambiar algunos nombres de macros de errores en los archivos .h porque estaban declarados con diferente nombre en las librerías correspondientes de stm32h7xx_hal.

En el código se obtienen ángulos de Euler o Cuaterniones con las siguientes funciones

```
bno055_vector_t BNO055_EulerVector = bno055_getVectorEuler();
```

```
bno055_vector_t BNO055_EulerVector = bno055_getVectorQuaternion();
```

Y son puestos en una estructura con la siguiente forma (w = 0 en ángulos de euler):

```
typedef struct {
 double w;
 double x;
 double y;
 double z;
} bno055_vector_t;
```

Kinematics (PENDIENTE REDACTAR LIVESCRIPT)

Holonomic: movement constraints can be expressed solely in terms of position, if components move by same net amounts, final configuration is the same. Configuration space is known and independent of path taken.

Nonholonomic: constraints on velocity, e.g. a differential car cannot have “sideways” sliding velocity.

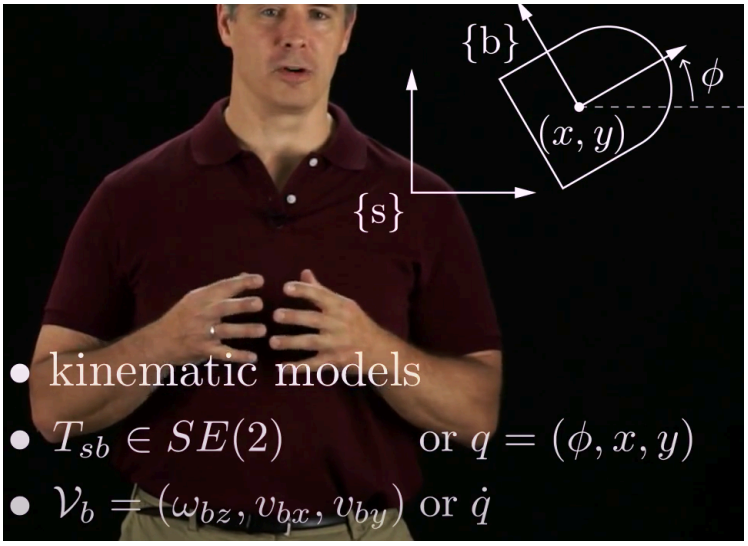
Modern Robotics, Chapter 2.4: Configuration and Velocity Constraints

Full Book MODERN ROBOTICS: MECHANICS, PLANNING, AND CONTROL

https://hades.mech.northwestern.edu/images/7/7f/MR.pdf

Chapter 13 Wheeled Mobile Robots (from same book ^^^)
https://modernrobotics.northwestern.edu/nu-gm-book-resource/13-1-wheeled-mobile-robots/#department
https://modernrobotics.northwestern.edu/nu-gm-book-resource/13-3-2-controllability-of-wheeled-mobile-robots-part-1-of-4/#department

Analysis (PENDIENTE REDACTAR BIEN LO HECHO EN MATLAB)



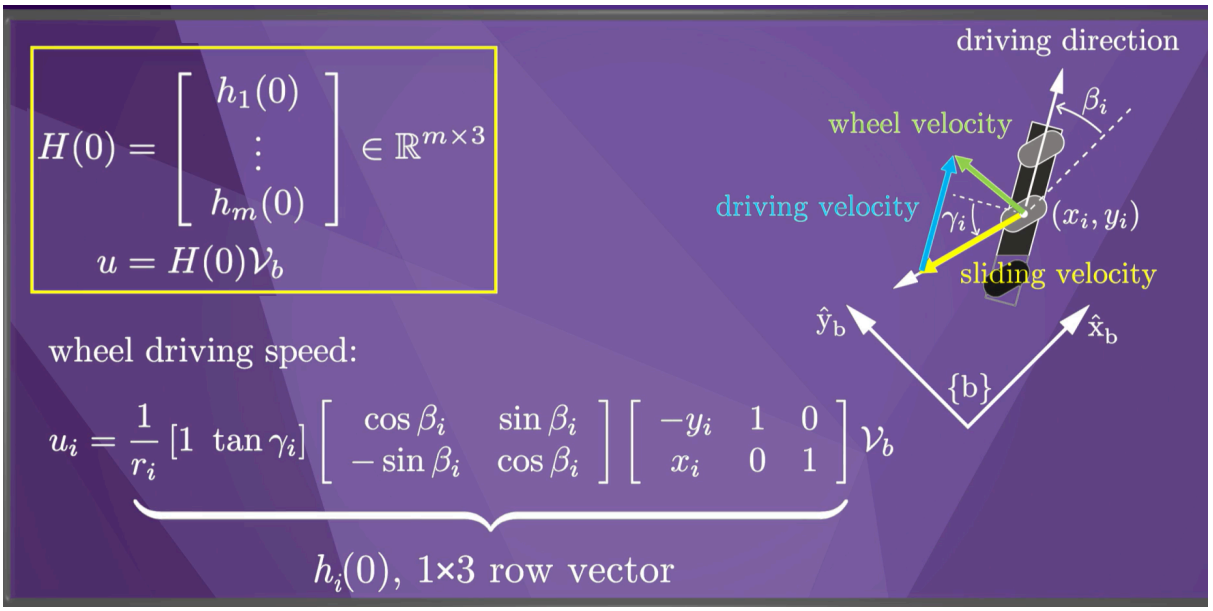
Se tiene un marco global {s} y un marco local {b}
La configuración del robot está dada por su posición y ángulo respecto al marco global

$$q = (\phi,\ x,\ y)$$

y su velocidad está dada por:

$$\dot{q} = (\dot{\phi}, \dot{x}, \dot{y})$$

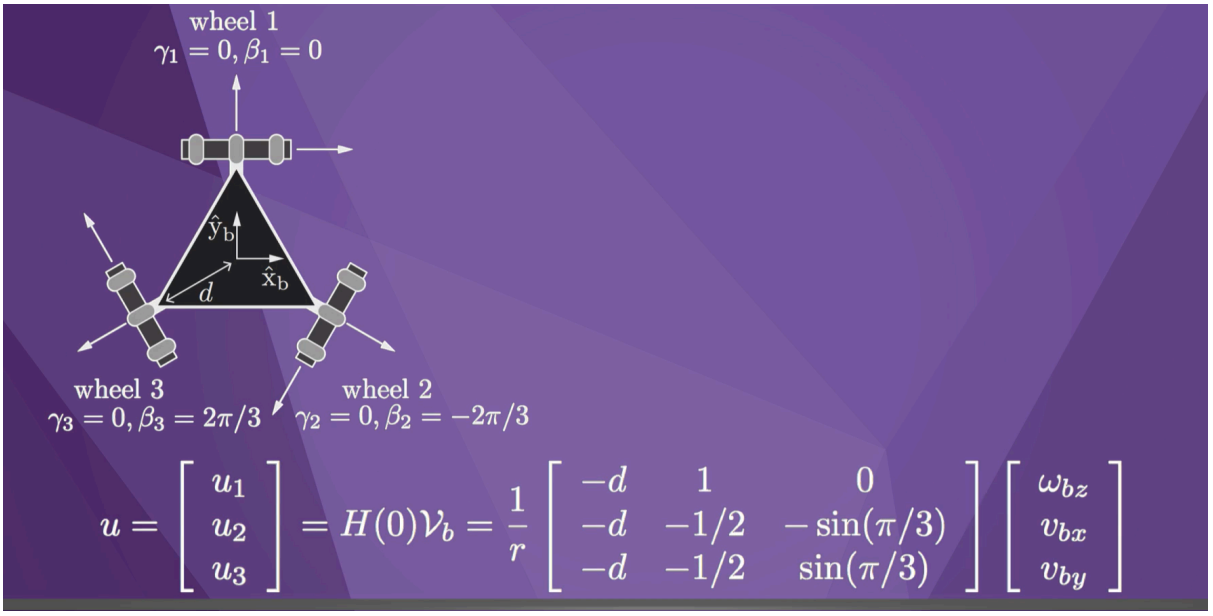
Given a desired chassis velocity, what should the driving speeds of the wheels be?



para cualquier llanta i

- Se tiene un centro de la llanta en una posición de (x_i, y_i) respecto al marco local $\{b\}$
- La dirección en la que gira la llanta está a un ángulo β_i respecto a los ejes (x_i, y_i)
- Los rodillos permiten movimiento a un ángulo γ_i respecto a la dirección $\perp$ a β_i. para una llanta omni $\gamma_i = 0$

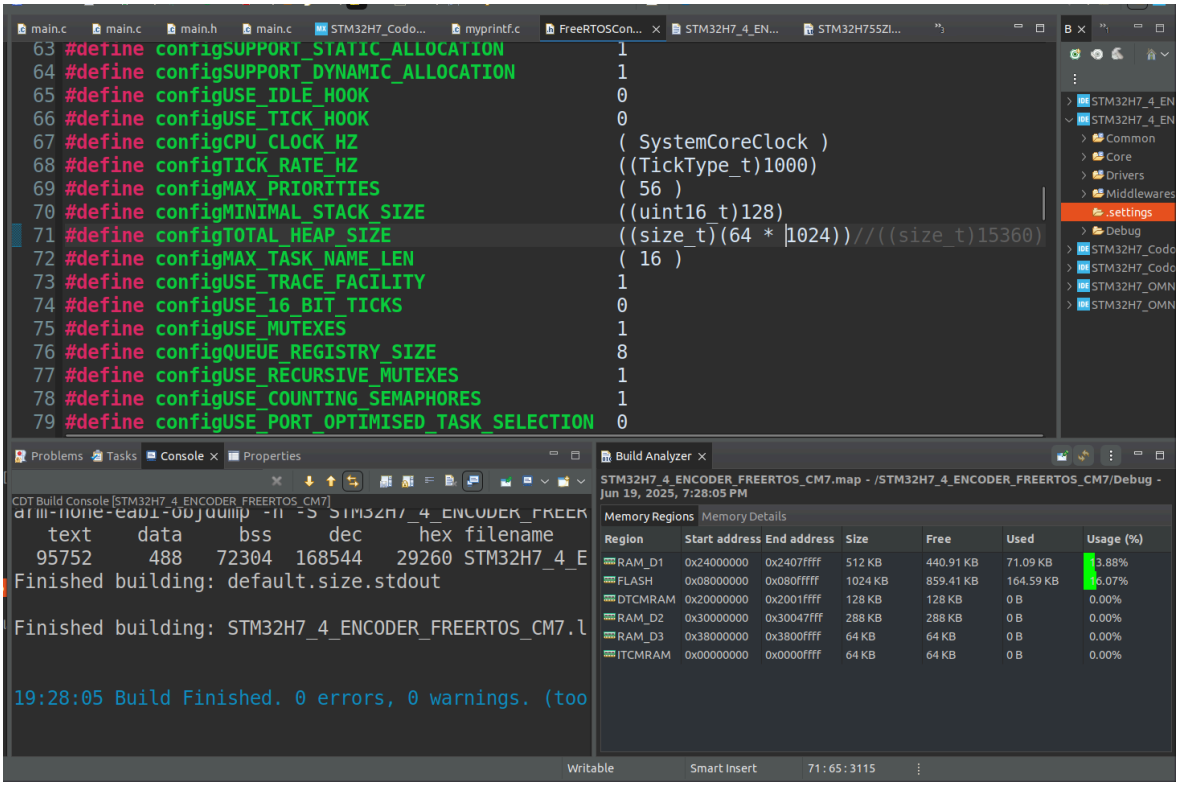
En base a esas definiciones se puede calcular la velocidad de rotación del motor



https://repository.umng.edu.co/server/api/core/bitstreams/97d7a947-b398-4d86-b3d6-63986f75ef82/content

Memory usage investigation

Se investigó sobre el uso de memoria de la stm usando freertos, pude ver que estaba por default en 15KB total de heap para todas las tasks.
Aparentemente esto se declara en la sección de .bss (cosas no inicializadas, creo q puestas con 0 por default) del archivo .ld que define el uso de memoria de la stm. La sección .bss está puesta dentro de la memoria RAM_D1.
Investigando vi que puedo ver el uso de memoria con el build analyzer abriendo el archivo .map que se genera cada que se buildea un proyecto. Dado que no parecía estarse usando mucha memoria del RAM_D1, le modifiqué el heap size de FreeRTOSConfig.h a que fuera 64KB, esto esto debería evitar problemas con el stack por un tiempo, ya que ahora es bastante más.
Si fuese necesario usar más stack, podría aumentarse el tamaño del heap en FreeRTOSConfig.h y/o cambiar el heap_4.c por el heap_5.c del repositorio oficial de FreeRTOS para que el heap sea declarado utilizando secciones de memoria diferentes en diferentes memorias físicas (RAM_D1, RAM_D2, RAM_D3). Sería necesario investigar a profundidad cómo hacer bien esto si fuese necesario.



Decent explanation on Static memory, Heap and Stack:
Introduction to RTOS Part 4 - Memory Management | Digi-Key Electronics

Useful forum:
https://forums.freertos.org/t/adjusting-heap-and-stack-size/12657

Dualshock4 PS4 Bluetooth connection

La idea es conectar el Dualshock4 por bluetooth a un ESP32 y mandar la información del ESP32 por uart a la stm32.

Troubleshooting

1. **TAMAÑO DE STACK:** si alguna vez aparecen errores de inicialización de i2c o simplemente no jala nada, puede ser por el tamaño del stack de alguna task, que se llenó, me pasó muchas veces.

Resources

Núcleo STM32H755 Documentation

Datasheet: https://www.st.com/resource/en/datasheet/stm32h755bi.pdf

Reference Manual:
https://www.st.com/resource/en/reference_manual/rm0399-stm32h745755-and-stm32h747757-advanced-armbased-32bit-mcus-stmicroelectronics.pdf

All STM32H7 docs:
https://www.st.com/en/microcontrollers-microprocessors/stm32h745-755/documentation.html?utm_source=chatgpt.com

God YT channels:
https://www.youtube.com/@stm32world/videos
https://www.youtube.com/@ControllersTech

MODERN ROBOTICS BOOK:
https://hades.mech.northwestern.edu/images/7/7f/MR.pdf

Copia de InClassSPICANbus_AD2024
Copia de QE_InClass_CANbus

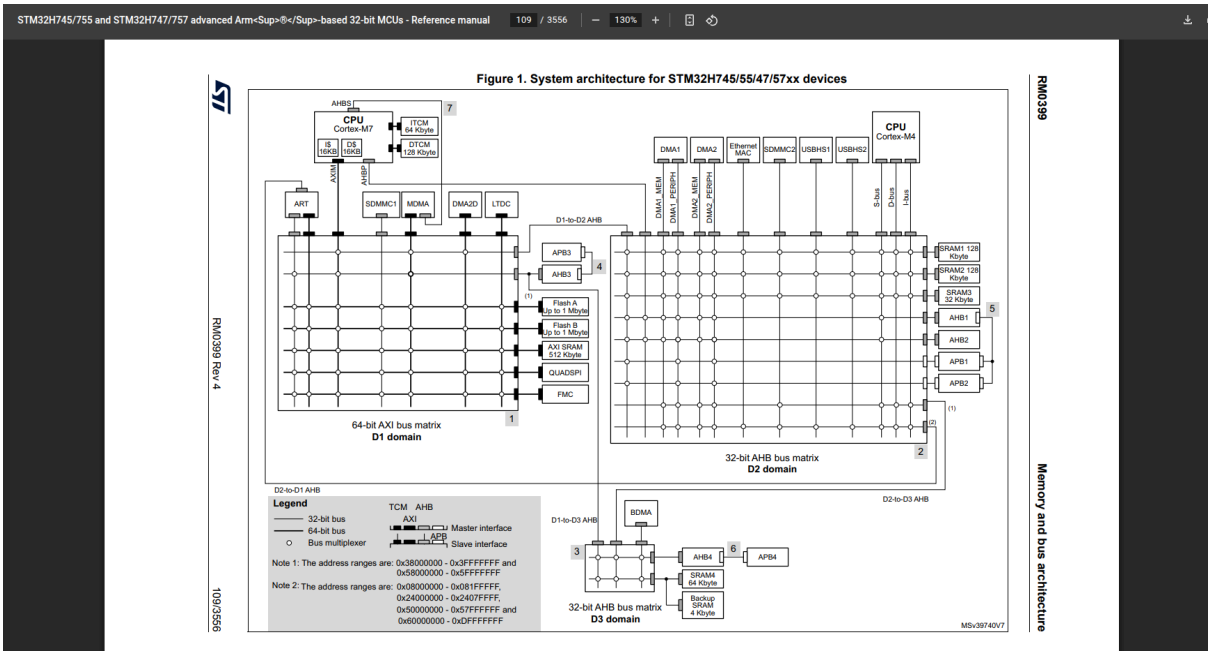
TJA1051:
https://pdf.utmel.com/r/datasheets/nxpusainc-tja1051te118-datasheets-2574.pdf

SCANF y PRINTF

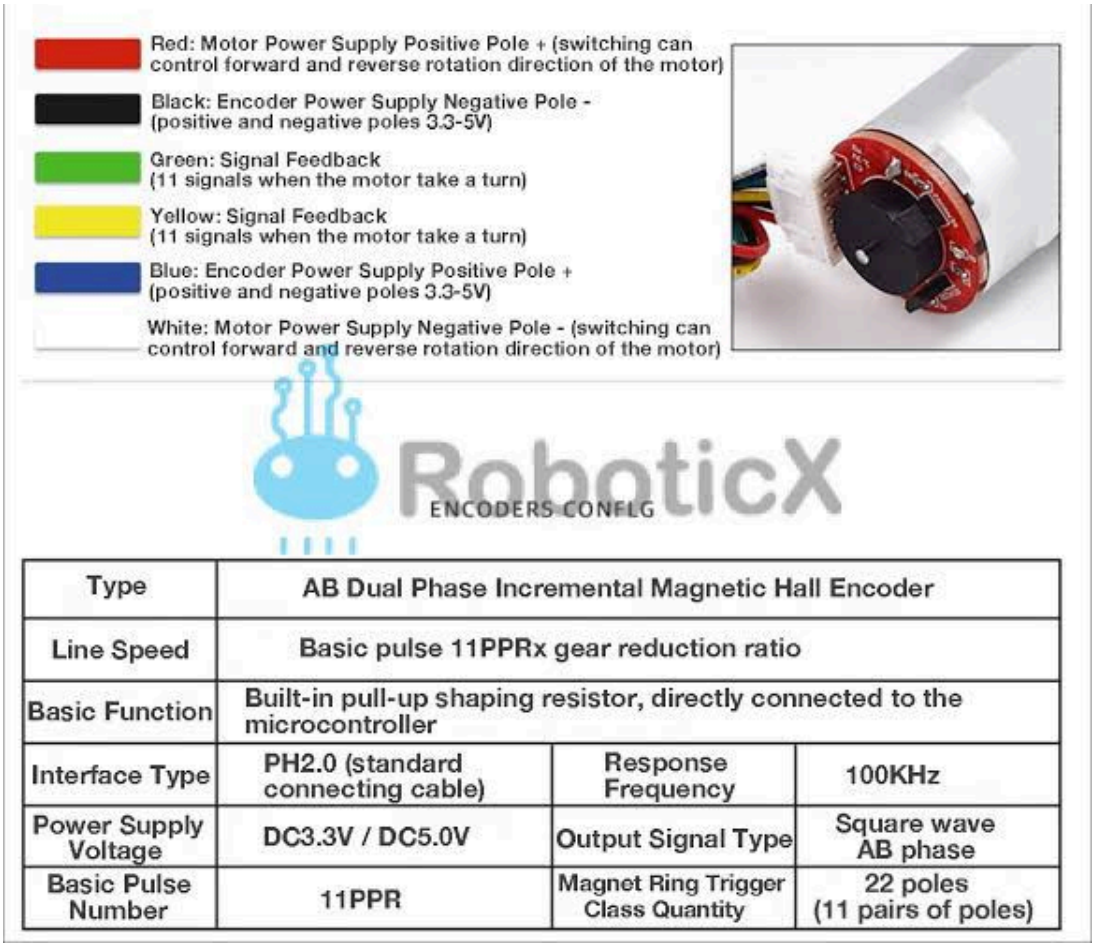
para ambos se deben activar las opciones de la sección properties → build → settings
Para printf basta con reescribir la función write en myprintf.h y myprintf.c
Para scanf también es lo mismo, pero se debe agregar esta linea al inicializar (después de hal_init() jala):

```
setvbuf(stdin, NULL, _IONBF, 0);
```

System Architecture



Motor con encoder en Chaneque

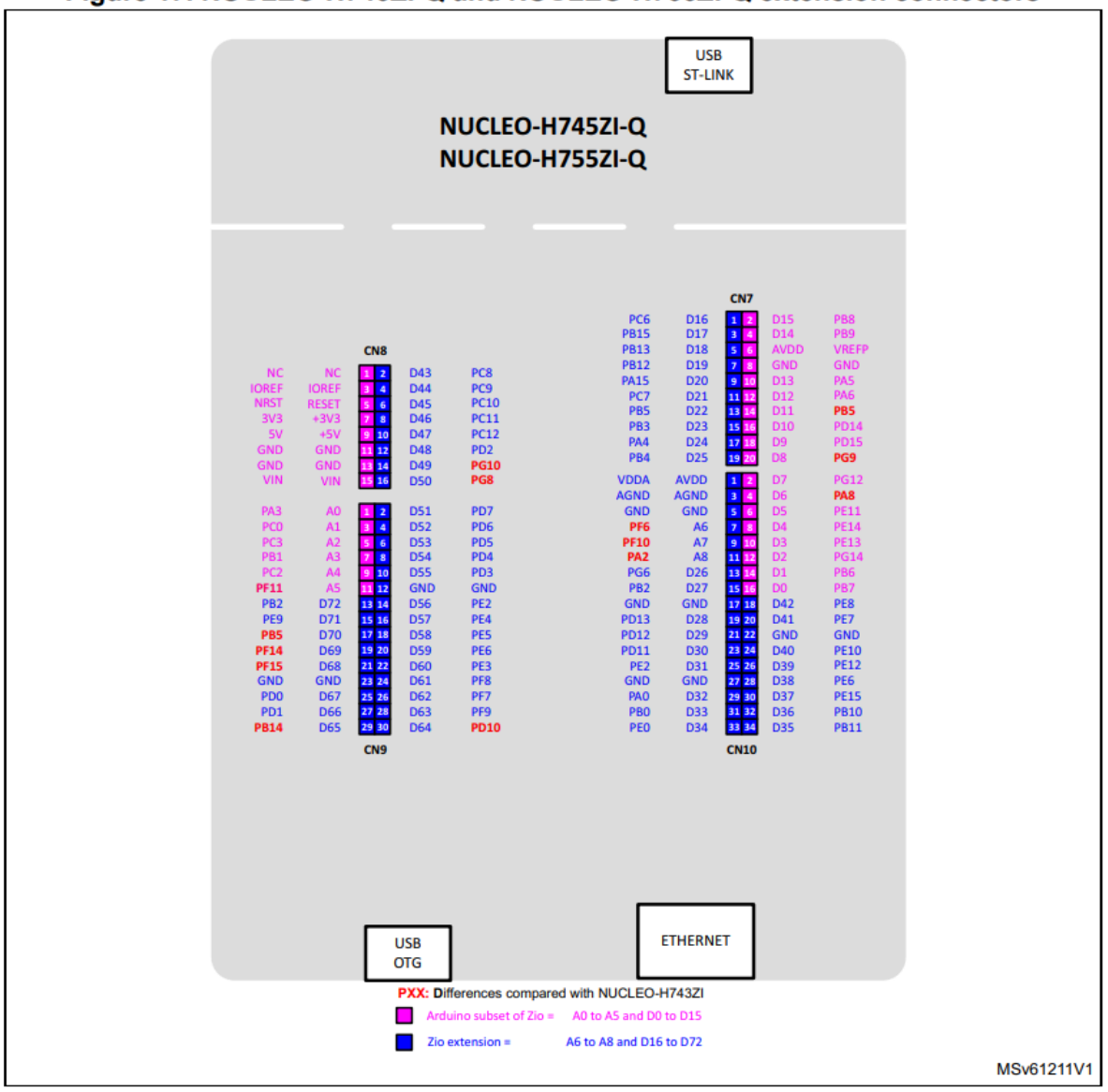


Type	AB Dual Phase Incremental Magnetic Hall Encoder		
Line Speed	Basic pulse 11PPRx gear reduction ratio		
Basic Function	Built-in pull-up shaping resistor, directly connected to the microcontroller		
Interface Type	PH2.0 (standard connecting cable)	Response Frequency	100KHz
Power Supply Voltage	DC3.3V / DC5.0V	Output Signal Type	Square wave AB phase
Basic Pulse Number	11PPR	Magnet Ring Trigger Class Quantity	22 poles (11 pairs of poles)

https://roboticx.ps/product/dc-gear-motor-with-encoder/

Female Headers Pinout

Figure 17. NUCLEO-H745ZI-Q and NUCLEO-H755ZI-Q extension connectors



UART Configuration

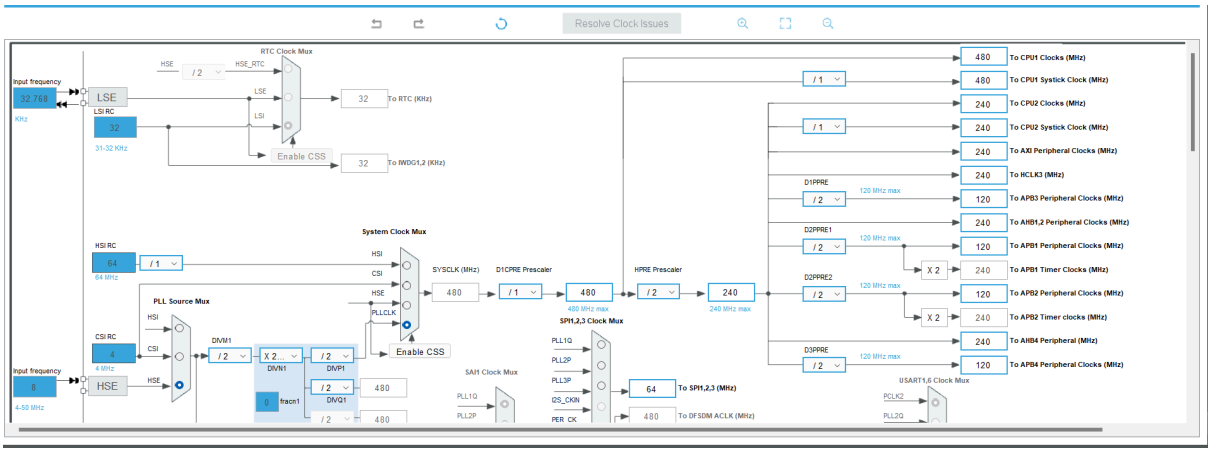
Note: the baud rate can be changed arbitrarily, as long as both ends have the same baud rate.

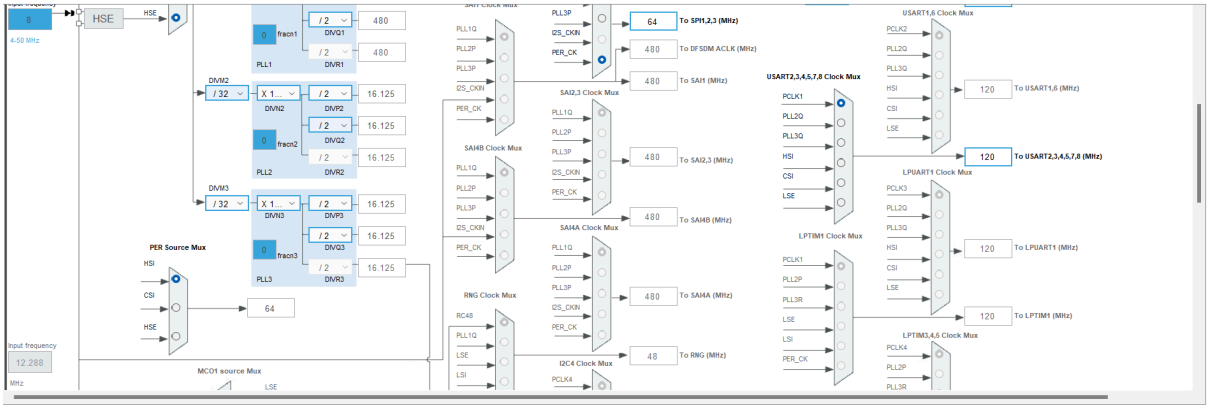
Pin Name	Signal on Pin	Pin Contex...	GPIO outp...	GPIO mode	GPIO P
PD8	USART3_TX	ARM Cortex-...	n/a	Alternate Fu...	No pull-u
PD9	USART3_RX	ARM Cortex-...	n/a	Alternate Fu...	No pull-u

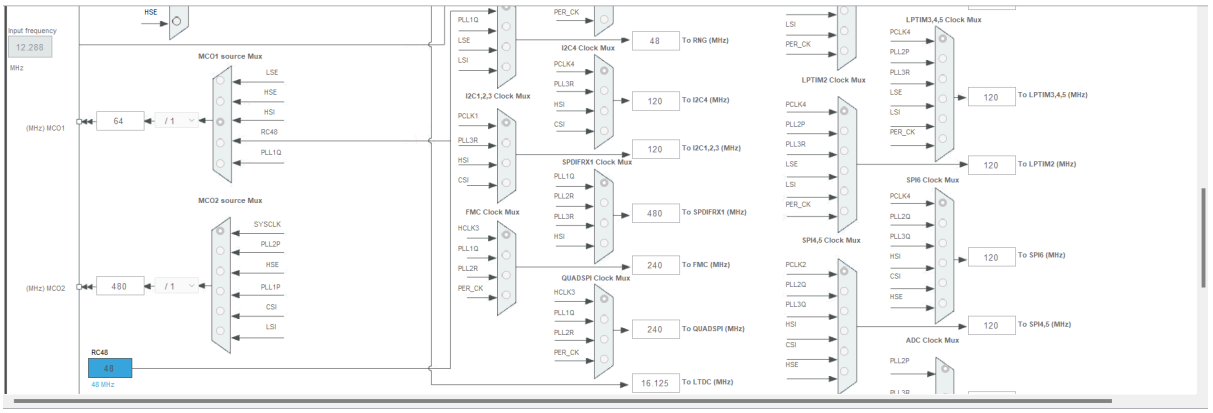
Uart polling and interrupt

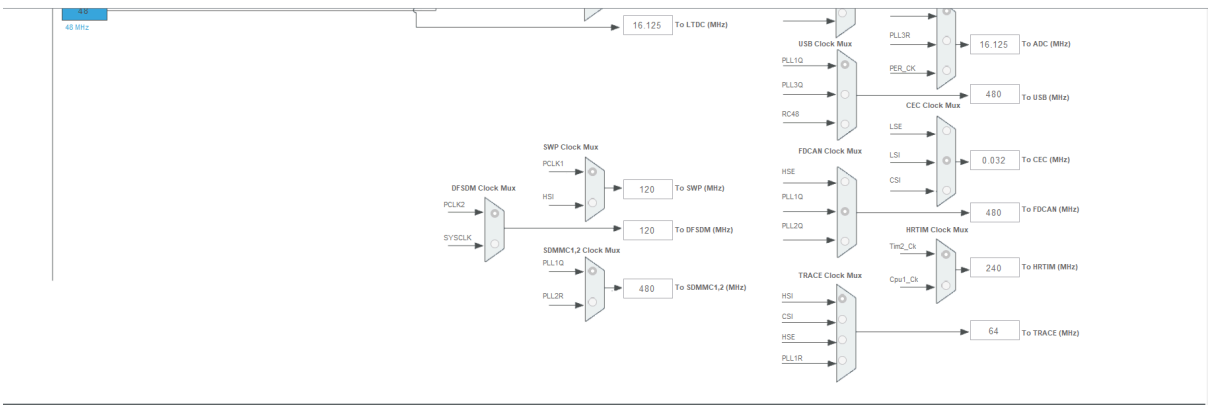
https://deepbluembedded.com/how-to-receive-uart-serial-data-with-stm32-dma-interrupt-polling/
https://controllerstech.com/stm32-uart-3-receive-data-in-blocking-interrupt-mode/

Clock Configuration







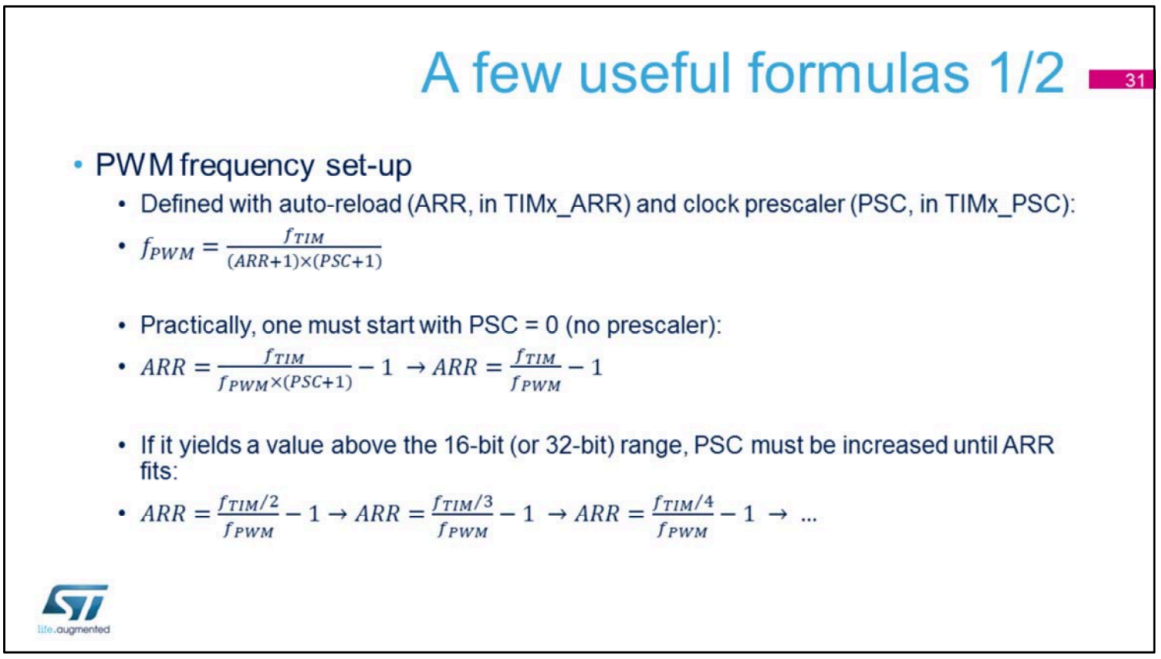


TIMERS

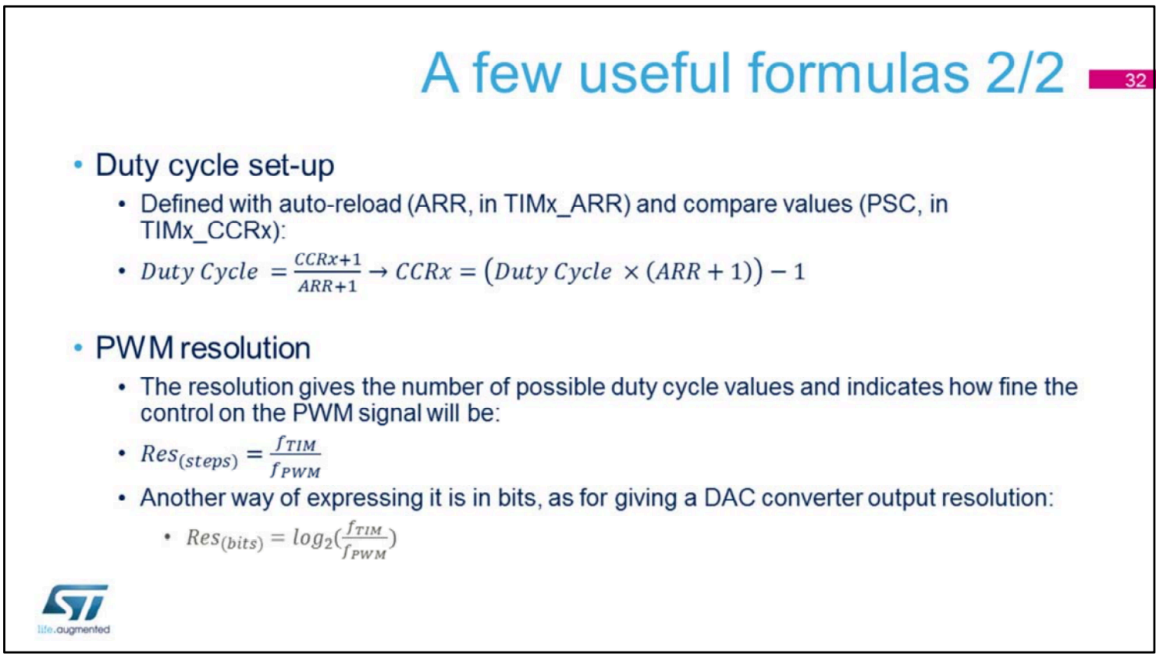
PWM

https://www.st.com/content/ccc/resource/training/technical/product_training/c4/1b/56/83/3a/a1/47/64/STM32L4_WDG_TIMERS_GPTIM.pdf/files/STM32L4_WDG_TIMERS_GPTIM.pdf/jcr:content/translations/en.STM32L4_WDG_TIMERS_GPTIM.pdf

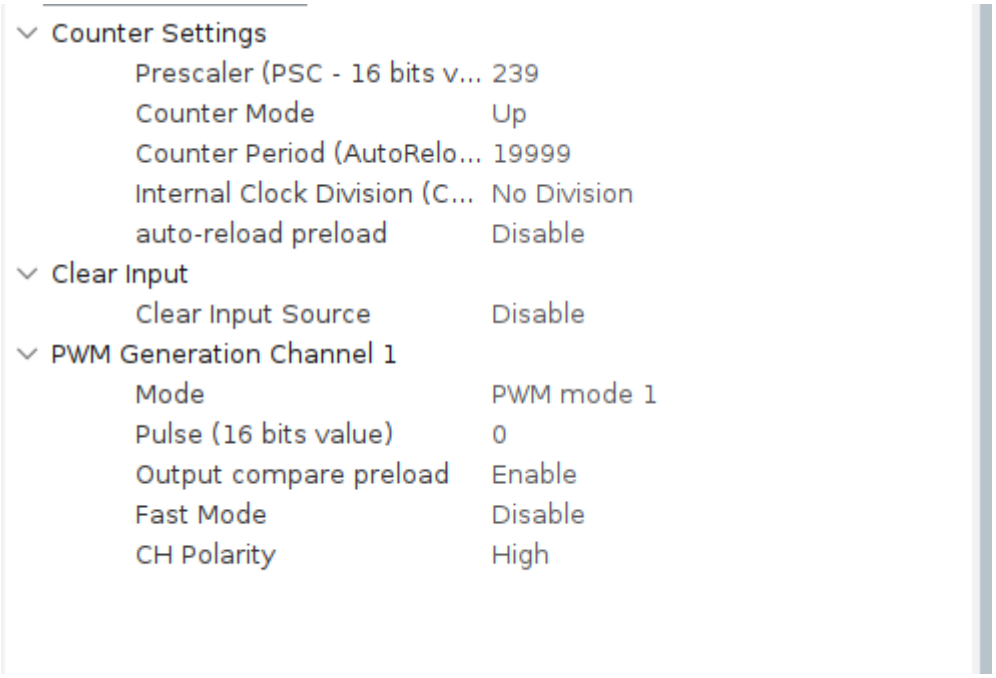
Pag 54:



Pag 56:



50 Hz example



Encoder mode